

ISABELLE STRATTON

LinkedIn

EXPERIENCE

Carnegie Mellon University Research Assistant	2026–Present Pittsburgh, PA
- Design and prototype sustainable materials from biomaterials and construction waste within a circular-economy framework. - Support research in material ecologies, exploring bio-based and waste-derived material systems. - Conduct material characterization and testing to evaluate performance and environmental impact.	
The University of Florida GIS Analyst	2025–2026 Gainesville, FL
- Built an ArcGIS tool that maps parcel-level ecosystem service values for conservation planning. - Processed spatial datasets (parcels, canopy, wetlands, soils) with reprojection, zonal statistics and field calculations. - Integrated economic valuation models with GIS layers for automated per-parcel outputs; developed a unified geodatabase and an interactive web map for county stakeholders. - Worked with Dr. Kotryna Klizentyte to develop a parcel-level ecosystem service value map for decision making.	
Behavioral Health Services North Data Analyst	2024–Present Plattsburgh, NY
- Deliver diagnostic and predictive analyses of healthcare needs aligned with CCBHC reporting standards. - Engineered a risk-prediction model using multivariate statistics and a Random Forest classifier. - Manage multiple databases in Excel and SQL and perform analytics on protected health information in full HIPAA compliance.	
Hulcr Lab Laboratory Technician	2023 Gainesville, FL
- Worked with Christopher Marais on PCA and K-means clustering for an AI-based bark beetle classifier. - Trained new technicians in sample processing and photography; operated a macro-camera for AI data collection.	

EDUCATION

Carnegie Mellon University	M.S. Sustainable and Regenerative Design	May 2027
University of Florida	B.S. Mathematics	August 2023
University of Florida	B.F.A. Fine Arts	August 2023

TECHNICAL SKILLS

Building analysis	Revit (modeling and energy analysis), IESVE (CFD and energy modelling), EnergyPlus, eQUEST, ClimateStudio, Climate Consultant
Computational design	Grasshopper, Python (parametric workflows and scripting)
3D modeling	Rhino, SolidWorks
Geospatial	ArcGIS ecosystem
Visualization	Adobe Creative Suite, Microsoft Office

PUBLICATIONS

Decision-making tool to value ecosystem services on potential land acquisitions in Hillsborough County, Florida Isabelle C. Stratton, Kotryna Klizentyte	In preparation Gainesville, FL
Publication in the works for the Hillsborough County ecosystem service valuation project; to be presented at the Tampa Bay Innovation Summit.	

Progress in Developing a Bark Beetle Identification Tool

2025

G. Christopher Marais, Isabelle C. Stratton, Andrew J. Johnson, Jiri Hulcr

Gainesville, FL

- Presents a tool for identifying bark beetles, invasive species that damage forests globally and whose uniform, homoplastic morphology makes identification difficult.
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PROJECTS

Generational Relief in Prosthetics

2023

- Was one of six in the 3D-Printed Assistive Devices Division; converted digital renderings of assistive prosthetics into 3D models for recipients.
- Troubleshoot and repaired 3D printers and helped adapt toys for movement-restricted children.

Touch-Activated Warming Sandbox

2022

- Wrote Arduino code for touch sensors and 12-volt heating elements, resolved voltage-regulation issues, and built a working prototype for educational and therapeutic use.
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AWARDS

- **Robert Long Mathematics Essay Contest, Honorable Mention** for an essay on a-periodic tilings in Islamic mosques (2023)
- **Neilson Scholarship**, with a preparator internship at the Harn Museum of Art; one of the youngest recipients (2020)
- **American Visions Award**, Scholastic Art and Writing, for illustration; one awarded per region nationally (2019)
- **National Gold Medalist**, Scholastic Art and Writing, illustration and mixed media (2018, 2019)
- **Bright Futures Scholar**, full undergraduate tuition (2019–2023)
- **Raimondi Family Photography Scholarship** (2019)
- **Jason Budjinski Memorial Scholarship**, academic excellence (2019)